

AVR BOD detects brownout conditions accurately at 1.8V, 2.7V and 4.5V within 2 μ s but draws about 20 μ A, adding substantially to the power-down current. But as part of the picoPower technology, it can be turned off to save power in extended standby, standby, power-save and power-down modes, as there is no need for the BOD while the MCU is in deep-sleep mode.

The picoPower BOD-disable feature is enabled by the application. It is disabled after the MCU enters the sleep mode, and enabled to verify that the power supply is sufficient before the MCU is allowed to wake up from sleep. The disable feature completely eliminates any power consumption penalty for the BOD during sleep mode and the MCU has full protection in active mode.

Ultra-low-power 32kHz crystal oscillator design utilised in AVR MCUs reduces current consumption in power-save mode to a level comparable to power-down mode's. With a supply voltage of 1.8 volts, the AVR picoPower technology is said to achieve the industry's lowest power-save current consumption of 650 nA with the 32kHz oscillator running and a sleeping BOD.

AVR also provides a method for atomic transition to the sleep mode. The SEI-SLEEP pair of instructions is guaranteed to execute atomically, most likely due to the AVR pipeline structure.

HC08/HCS08. HC(S)08 is an 8-bit CISC MCU family from Freescale Semiconductors. HC08 provides two LPMs (wait and stop), while HCS08 has two new stop modes (Stop1 and Stop2).

In wait mode, clocks to the CPU are turned off. All other peripherals can be enabled. Any interrupt can be used to exit wait.

Low-voltage detect (LVD) is enabled and states of all I/O pins are latched in stop mode. Individual clock generators are enabled but the clock to the rest of the MCU is turned

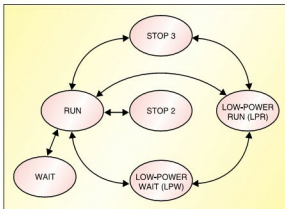


Fig. 10: HC08/ColdFire CPU mode transition

off to avoid long oscillator-startup times. Interrupt sources such as IRQ, reset, RTC and ADC are used to exit stop mode.

Stop1 is a full power-down mode, whereas Stop2 provides standby currents that are lower than Stop but higher than Stop1. Stop2 is a partial power-down mode. IRQ or reset is used to exit either Stop1 or Stop2 mode. Typical consumption in power-down mode is 20 nA.

HC08 devices are members of the successful HC08 family, having true 1.8V operation support without sacrificing performance. These have a 16MHz CPU/8MHz bus at 1.8V. High integration includes four serial communication ports, up to eight timer/PWMs, and an 8-channel 10-bit ADC specified down to 1.8V.

The ADC continues normal operation during wait mode. Any enabled-CPU interrupt request from the ADC can bring the MCU out of wait mode. It can be powered down if the ADC is not required to bring the MCU out of wait mode. But it is inactive when the MCU enters stop mode. Any pending conversion is aborted. ADC conversion resumes when the MCU exits stop mode. One conversion cycle is needed to stabilise the analogue circuitry before using ADC data after exiting stop mode. CPU clock is disabled in LPMs and it begins running after the oscillator stabilisation delay.

The HC(S)08 instruction set in-

cludes special instructions—wait and stop—for transition to low-power wait and stop modes, respectively. The HC08 documentation states very clearly that both wait and stop instructions atomically enable interrupts as a side-effect of entering the sleep mode. The IRQ module remains active in wait and stop modes. Clearly, the HC08 datasheet anticipated that

transition to the LPM must happen with interrupts disabled.

Flexis MCU (ColdFireVx). QE128, a ColdFireV1, is a 32-bit variable-length RISC core. QE128 devices have five LPMs. Low-power run (LPR) mode is the equivalent of a low-speed active mode with the frequency-lock loop (FLL) disabled. The MCU can be put into and out of LPR mode.

In wait mode, the CPU is not clocked but the FLL is enabled. In low-power wait (LPW) mode, the CPU is not clocked and the FLL is disabled. Exit from wait and LPW can be achieved through any enabled interrupt. Therefore any module can place the MCU back into run or LPR mode.

There is a clear map of allowable mode transitions. Entry into Stop3 and Stop2 is governed by setting the background debug mode and LVD modules. Any module as well as a reset can bring the MCU out of Stop3. Stop3 has wakeup time of 6 μ s. RTC, IRQ or RESET are the only paths out of Stop2. Stop2 mode exits as if a power-on reset was executed.

Unlike the ADC of MSP430, which can be triggered from a timer event, the ADC of QE128 can be triggered from the RTC. The difference is the timer trigger, which is more flexible (input capture, output compare or PWM), but QE128's RTC trigger is periodic by nature. The ADC module of MSP430 operates down to 2.2V, whereas the ADC of QE128 continues